INCH-POUND

MIL-P-50357 (AR)
AMENDMENT 3
15 August 1994
SUPERSEDING
AMENDMENT 2
30 September 1974

MILITARY SPECIFICATION

PROJECTILE, 60MM H.E., M49A3
METAL PARTS ASSEMBLY (ALLOY STEEL)

This Amendment forms a part of MIL-P-50357 (AR), dated 30 September 1969, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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* 2.1, Delete "MIL-A-2550....." in its entirety.

Delete "MIL-I-6868....." in its entirety.

Add "MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specifications For"

Under heading "DRAWINGS" delete "Munitions Command" and substitute "U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)."

Beneath the heading described above add:

"7548319 - Box, Fiberboard Ammunition for Interplant Shipment of Projectiles, Empty"

Delete "MIL-STD-109....." in its entirety.

Delete "MIL-STD-1235....." in its entirety.

After "MIL-STD-1169....." add,

"MIL-STD-1949 - Inspection, Magnetic Particle"

Delete "Publications .. PA-PD-792 ..Equipment, Testing or Measuring for Ammunition Items"

AMSC N/A **FSC 1310**

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* 2.2, Delete "ASA-B.46.1" and substitute "ANSI B46.1".

Delete "(Application for copies...)" in its entirety and substitute, "(Application for copies should be addressed to: American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)"

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* Add new paragraph 3.3.2 as follows:

"3.3.2 Hydrostatic test. Each projectile body shall be able to withstand a hydrostatic test load of 7500 +500 psi without cracking, leakage, rupture or permanent deformation when tested in accordance with 4.3.3.1.4."

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* 3.7.2, Delete "and shall be in accordance with the provisions of MIL-A-2550."

4.1, Add to the end of the paragraph:

"Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification."

* 4.1.1, Delete in its entirety and substitute:

"4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material."

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* 4.1.2, Delete in its entirety.

* 4.2, Delete section in its entirety and substitute:

"4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.2.1 First article inspection. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of MIL-A-48078, 4.3.2.2, 4.3.2.3, 4.3.3.1.2.1 and 4.3.4 herein. The first article sample shall consist of:

Four tensile specimens (total) from two bodies (see 4.3.3.1.2.1), ten unpainted bodies (see 4.3.2.2) and thirty painted body assemblies. Five of the painted assemblies shall be subjected to the salt spray test and the remainder shall be inspected in accordance with 4.3.2.3 and forwarded for the proving ground tests.

4.2.2 Rejection. See MIL-A-48078."

* 4.3, Delete section in its entirety and substitute:

"4.3. Quality conformance inspection.

4.3.1 Lots

4.3.1.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following, as applicable:"

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* 4.3.2.1, Delete in entirety and substitute:

"4.3.2.1 Tool control dimension. Dimension marked tool control shall be gaged during First Article testing and after each 250 items produced on each machine in operation and after any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor, whenever a defective tool control dimension is found and all material produced since the last successful inspection shall be rejected."

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* 4.3.2.2, Critical 1, delete in its entirety and substitute new critical "1 ...Hydrostatic test ...Gage/4.3.3.1.4".

Delete Minor 202 in its entirety.

PAGE 8

* 4.3.2.3, Critical 1, delete "(probe light)".

Add "Special a ...Hydrostatic test stamp missing ...100% ...Visual".

Delete Major 103 in its entirety and substitute:

"103 ...Minor diameter of large thread, max. for min. length ...Gage"

Delete Major 108 in its entirety.

Add Major 114:

"114 ...Length of minor diameter of large thread, max. ...Gage"

* 4.3.2.4, after "Packed carton" add: "Drawing 7548319".

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4.3.2.5, after "Packed carton" add: "Drawing 7548319".

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* Add new paragraph 4.3.3.1.4 as follows:

"4.3.3.1.4 Hydrostatic test. After final machining, each projectile shall be hydrostatically tested to determine compliance with 3.3.2. The equipment used shall be capable of performing the test accurately and consistently. During the test, the pressure will be applied until the required pressure is reached and held for five (5) seconds, minimum. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over four (4) hours of continuous operation. It shall also be calibrated prior to restarting the operation after a layoff of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower than the minimum requirement, the bodies tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. Projectile rupture, cracking, leakage, or bulging during the test shall be cause for rejection of the body. The test fixture shall not restrict the projectile body from expanding longitudinally or radially under pressure. Protective barriers should be installed to shield operators from metal fragments which may come from defective projectiles that fail the test. If threads are engaged, the hydrotest plug should be designed with a thread relief, to prevent loading of initial threads on the projectile. Examination should be made of all hydrotest plugs after a projectile is blown during testing. Badly nicked or damaged plugs may cause destruction of acceptable items."

* 4.3.3.2.1.a, Delete "affect" and substitute "effect".

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4.3.4, Delete section in its entirety and substitute the following:

* "4.3.4 Proving ground tests.

4.3.4.1 Maximum average pressure test. (See 4.4.4.1) Five (5) projectiles shall be tested from the First Article Sample, from each of the first three (3) production lots and from three (3) consecutive lots when a new mill heat of steel is introduced into production. However, any lot may be subjected to this test, retested, or testing may be reinstituted when so directed by the procuring contracting officer.

4.3.4.2 Service pressure test. (See 4.4.4.2) Twenty (20) projectiles shall be tested from the First Article sample, from each of the first three (3) production lots, and from three (3) consecutive lots when a new mill heat of steel is introduced into production. However, any lot may be subjected to this test, retested, or testing may be reinstituted when so directed by the procuring contracting officer."

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* 4.3.5, Add "See 6.4" at the end of the paragraph.

PAGE 13

4.4.1.2, Second line, change "Four Tension Test Specimens" to "Two Tension Test Specimens."

PAGE 14

* 4.4.4.2, First sentence, delete "for comparison with an equal number of rounds from a reference lot".

Second sentence, delete in its entirety.

* 5.1, At the end of the sentence add "and Drawing 7548319".

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* 6.4, Add new paragraph as follows:

"6.4 Submission of inspection equipment designs for approval. See paragraph 6.2.3 of MIL-A-48078. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center (ARDEC), ATTN: SMCAR-QAT-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationships to the last previous amendment.

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